Video Corpus Moment Retrieval With Query-Specific Context Learning and Progressive Localization

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***Abstract*— Video corpus moment retrieval (VCMR) aims to retrieve a moment from a large corpus of untrimmed videos corresponding to a given language query. However, existing methods often fall short due to their reliance on simple cross-modal attention mechanisms and one-stop localization, which fail to handle the complex multimodal information and large search space effectively. To address these challenges, we propose a novel VCMR method with Query-specific Context Learning and Progressive Localization (QCLPL). First, we construct query-specific multimodal contexts that capture complementary and consistent semantics across subtitles and frames, ensuring informative and efficient context building. We further introduce a semantic contrastive loss to refine these multimodal contexts, filtering out query-irrelevant information. Additionally, we introduce a progressive localization strategy that transforms the moment localization task into a two-stage process. By classifying frames into foreground and background regions, we present a simplified binary classification problem before boundary prediction, constrained by a region-aware loss. This progressive approach leverages region priors to improve subsequent moment localization. Extensive experiments on the TVR and DiDeMo datasets demonstrate that our method significantly outperforms existing approaches, setting a new state of the art for VCMR.**

***Index Terms*— Video corpus moment retrieval, query-specific context learning, progressive localization.**

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I. Introduction

With the fast development in communication and media creation technologies, video has become a dominant medium for information transmission [1], [2], [3], [4], [5], [6]. Consequently, there is a growing demand for techniques capable of quickly retrieving video segments of user interest based on natural language. Single Video Moment Retrieval (SVMR) [7], [8], which aims to retrieve a specific moment from an untrimmed video that is semantically related to a given text query, has attracted great attention. However, current SVMR benchmarks typically assume the existence of a ground truth video containing target moment [7], [8]. This assumption limits the applicability of these methods in real-world scenarios where such ground truth is often unavailable. To bridge this gap, recent works have introduced the Video Corpus Moment Retrieval (VCMR) task [9], [10], which involves predicting the relevant moments corresponding to a text query from a large video corpus.

Despite the significant progress that has been made in SVMR, VCMR is far from mature due to the following challenges. (1) *Modal heterogeneity.* As shown in Fig. 1, to align with real-world scenarios, VCMR involves localizing moments based on complex multimodal information in the video, *i.e.*, textual subtitles and visual frames. Modal heterogeneity poses a fundamental challenge to capture consistent and complementary contexts from both modalities. (2) *Large search space.* Unlike SVMR, VCMR performs moment localization across a large corpus of videos rather than a single video. This significantly increases the search space, requiring the model to distinguish subtle differences between numerous moment proposals, thereby making accurate retrieval more difficult.

Most existing VCMR methods focus on the first challenge, *i.e.*, employing advanced fusion strategies to obtain multimodal context. Early efforts often adopt two scalar weights to fuse subtitle and frame streams [11], [12], [13]. As shown in Fig. 2 (a), they assign fixed weights over time to balance the influence of subtitles and frames, but ignore that only limited parts of the video are related to the query, resulting in redundant and query-irrelevant contexts. More recent works [14], [15] propose to apply cross-attention on queries and videos, as shown in Fig. 2 (b). Nevertheless, the complex scenario in an untrimmed video, *e.g.*, footage transitions and

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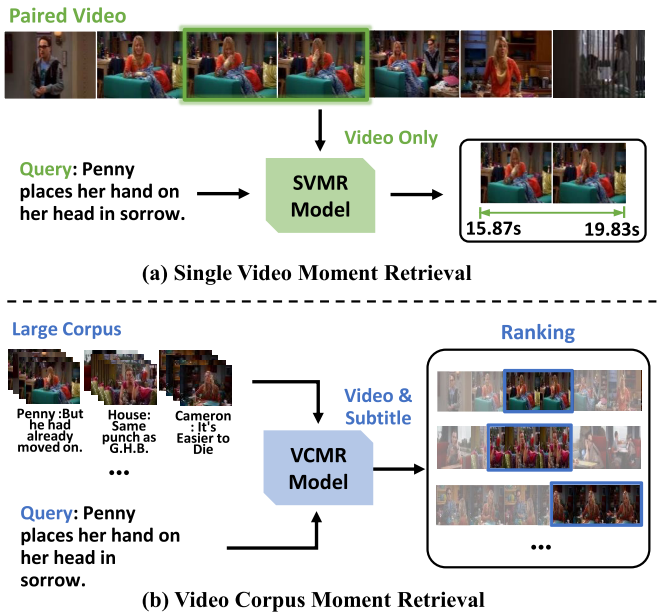

Fig. 1. Comparison of SVMR and VCMR tasks. VCMR presents two emerging challenges: modal heterogeneity in visual frames and textual subtitles, and large search space posed by the large corpus.

irrelevant events, will increase the likelihood of attention drift which leads to the undesirable local activation [16].

As for the second challenge, a common practice in VCMR is to divide the task into two stages: video retrieval and moment localization [11]. Initially, a text-to-video retriever [17] generates a set of candidate videos with relevance scores based on the query. The model then retrieves target moments from these candidates. Although the candidate set is smaller than the full corpus, the retrieval task remains challenging since all candidates are related to the query to varying degrees, making it difficult to distinguish the target moment. Existing methods [11], [12], [13], [14], [18] typically rely on a 1D convolutional layer to predict the start and end frames over all frames directly. These approaches leverage the natural property that clips inside a semantically close span have higher similarity to the query than those outside, forming detectable edges around the span. However, the indistinguishable retrieval candidates in VCMR make these edges less distinct, limiting the effectiveness of convolutional filters for moment localization.

To tackle the above issues, we propose a novel VCMR method with Query-specific Context Learning and Progressive Localization (QCLPL). At first, we make efforts to capture informative multimodal contexts that are modality-complementary and semantic-consistent. As shown in Fig. 2 (c), we decompose the original query into two modality-specific subqueries–one for subtitles and one for frames. Using these subqueries, we separately capture query-related portions from each modality. The extracted information from both modalities is expected to describe the same query consistently. To ensure this consistency, we devise a semantic contrastive loss $\mathcal{L}^{sc}$ to penalize the contexts that include query-unrelated semantics and ensure that both frame- and subtitle-based contexts are more closely aligned with the query than with negative videos. In this way, our multimodal context integrates the key clues pertinent to the query to mitigate attention drift, while discarding substantial redundant information. This refined context also has the potential advantage of improving retrieval efficiency.

Building on the query-specific multimodal contexts, we propose a progressive localization method to handle the problem of moment retrieval within a large search space. Specifically, we consider the localization process as a two-stage task. First, we introduce a pair of labels for the foreground and background regions, which presents a more manageable binary classification problem before the sparse boundary prediction. We label the frames within the ground truth moment and several neighboring frames as foreground, while the rest as background. By applying region classification, we learn a region embedding to guide the subsequent moment localization task. Unlike vanilla convolution-based methods [10], [11], [12], [13], [14], [19], this approach provides a progressive learning framework, facilitating boundary detection with the foreground and background region priors. We also introduce a region-aware loss $\mathcal{L}^{ra}$ to ensure the prediction of ground-truth foreground and background regions. Extensive studies on the benchmark datasets (*i.e.*, TVR and DiDeMo) demonstrate the efficacy of our method.

Our contribution can be summarized as follows:

- We present a novel approach to tackle the key challenges of modal heterogeneity and large retrieval space in VCMR, called Query-specific Context Learning and Progressive Localization (QCLPL).
- We introduce a semantic contrastive loss $\mathcal{L}^{sc}$ for VCMR to ensure modality-complementary and semantic-consistent multimodal contexts, and a region-aware loss $\mathcal{L}^{ra}$ to capture useful moment-relevant features by differentiating between foreground and background regions.
- Extensive experiments on two benchmark datasets, TVR and DiDeMo, demonstrate that our QCLPL achieves state-of-the-art performance over existing methods.

II. Related Work

A. Singe Video Moment Retrieval

Single video moment retrieval (SVMR), also known as natural language video localization (NLVL), aims to retrieve relevant moments from an untrimmed video that semantically corresponds to a given query. Current approaches in SVMR can be broadly categorized into two primary streams: proposal-based methods [20], [21] and proposal-free methods [22], [23], [24], [25], [26]. Additionally, reinforcement learning-based strategies [27], [28] and other supervised learning techniques [29], [30], [31], [32], [33], [34] have also been explored. Despite the progress, these methods often struggle with localization accuracy [35], particularly in overcoming the challenges of cross-modal feature fusion and modality imbalance between video content and textual queries. To this end, SeqPAN [24] introduces a self-guided parallel attention mechanism to capture intra-modal contexts and cross-modal interactions between videos and texts. CDS [22] tackles the visual and combinatorial bias commonly found in SVMR datasets, and presents a collaborative debiasing structure.

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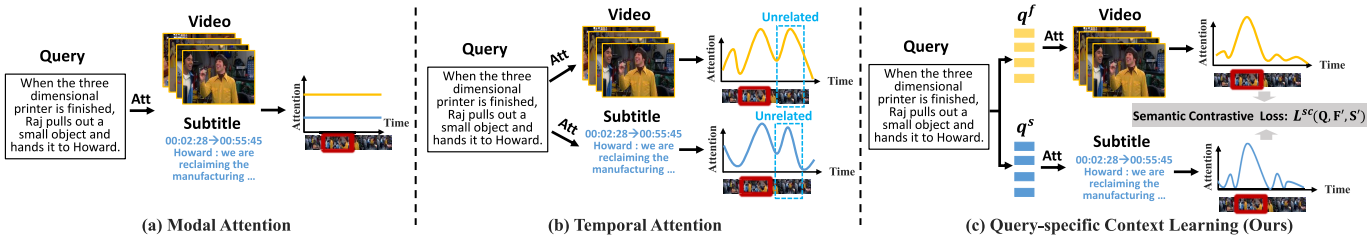

Fig. 2. Different multimodal fusion methods in VCMR. (a) Modal attention-based methods [11], [12], [13] use two scalar weights to fuse subtitle and frame streams. (b) Temporal attention-based methods [14], [15] rely solely on the cross-modal attention between queries and videos. (c) Our query-specific context learning method introduces a semantic contrastive constraint.

MCMN [23] employs momentum-based cross-modal contrast to generate precise and discriminative representations. Similarly, CRNet [21] emphasizes modeling multi-granularity cross-modal relations and temporal relations among video moments, enhancing SCMR performance. FMRN [36] further explores fine-grained modality relation-aware information within text queries, videos, and proposals.

B. Video Corpus Moment Retrieval

While SVMR has shown promising results, its assumption that the ground-truth video is pre-identified is impractical in real-world applications. To address this limitation, Escorcia et al. [9] introduced the Video Corpus Moment Retrieval (VCMR) task that aims to retrieve semantically relevant moments from untrimmed video corpus based on a given language query. Current VCMR approaches can be categorized into two groups: one-stage methods [10], [19], [37], [38] and two-stage methods [11], [12], [13], [14]. One-stage methods typically employ a single model to jointly handle both video retrieval and moment localization. XML [10] is the first model specifically designed for VCMR, introducing the TVR dataset and extending moment retrieval from a single modality (visual) to a multimodal framework (visual and subtitle). HAMMER [38] adopts a hierarchical fusion approach to integrate coarse-grained, fine-grained, and multimodal information. RelocNet [19] improves feature learning by employing contrastive learning at both the video and frame levels. In contrast, two-stage methods separate video retrieval and moment localization into distinct modules, leveraging the strengths of each. The representative models include CONQUER [11], DFMAT [13], CTDL [39] and CKCN [12]. DMFAT [13] focuses on VCMR by introducing a multi-granularity feature fusion approach with adversarial training. CTDL [39] presents a counterfactual two-stage debiasing learning to overcome the retrieval bias caused by spurious correlations between query and scene. These two-stage models consistently utilize a pre-trained HERO [37] as video retriever and an early-fusion moment localizer. In contrast, PREM [14] employs relevant content-enhanced contrastive learning to optimize the video retriever and to fine-tune HERO during training.

Most existing VCMR methods focus on developing advanced fusion strategies to obtain multi- modal context [11], [12], [13], [14]. They rely on fixed scalar weights [11], [12], [13] or cross-attention mechanisms [14] to integrate multimodal information in frames and subtitles, which often results in redundant and query-independent contexts. For moment localization, existing methods [11], [12], [13], [14], [19] commonly apply a 1D convolutional layer to predict the start and end timestamps directly, as proposed in [11]. These methods struggle with sub-optimal performance since they oversimplify the inherent challenges in VCMR posed by the large and hard retrieval space.

In this work, we propose a novel VCMR method with query-specific context learning and progressive localization, which aims to yield modality-complementary and semantic-consistent multimodal contexts while improving localization accuracy through a two-step refinement process.

C. Contrastive Learning in WVMR

This work leverages contrastive learning to address the challenges of extracting query-specific multimodal contexts. Contrastive learning has been widely explored in the field of weakly supervised video moment retrieval (WVMR) [30], [31], [32], [33], [34]. Unlike fully supervised settings, the start and end timestamps are inaccessible during training for WVMR. To overcome this challenge, several previous works [30], [31], [32], [33] have made significant progress. For instance, VCA [33] introduces a contrastive learning approach for learning the moment-level alignment of videos containing semantically related moments. CNM [32] introduces a contrastive negative sample mining strategy with an intra-video contrastive loss to make the positive and negative samples discriminative. Meanwhile, CPL [31] proposes a contrastive proposal learning method using multiple learnable Gaussian functions to generate both positive and negative proposals within the same video. GMP [30] enhances proposal diversity through Gaussian mixture proposals and employs contrastive learning to differentiate query-relevant locations from confusing negative proposals.

Our work is significantly different from these WVMR studies. We propose a semantic contrastive loss specifically tailored for the VCMR task, focusing on effectively extracting multimodal context from video frames and subtitles. This loss is used to penalize unrelated semantics and encourage query-relevant content in multimodal contexts.

III. Method

Given a corpus of untrimmed videos $\mathcal{V}$ and a language query q, the goal of the VCMR task is to retrieve the moment $m^* = (t^s, t^e)$ that is most relevant to the query q from $\mathcal{V}$, where t^s and t^e denote the start and end timestamps of the

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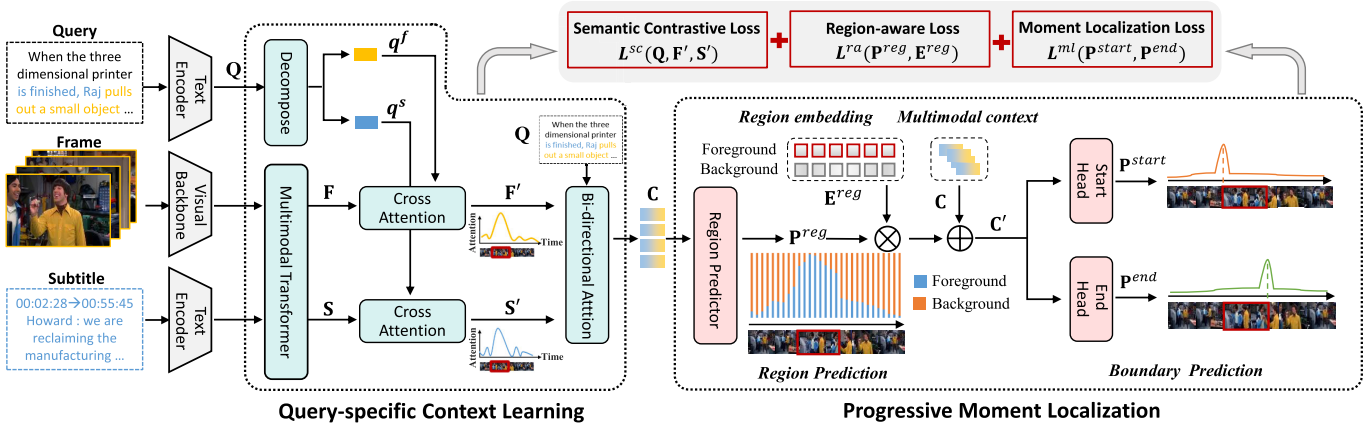

Fig. 3. Overview of our proposed QCLPL. It mainly consists of query-specific context learning and progressive moment localization modules. The former constructs the modality-complementary and semantic-consistent multimodal context $\mathbf{C}$ and the latter successively performs region prediction and boundary perdition to give the final moment span. The model is jointly optimized with multi-task losses, including a moment retrieval loss, a semantic contrastive loss, and a region-aware loss.

target moment, respectively. Formally,

$$m^* = \underset{m}{\operatorname{argmax}}\ P(m|q, \mathcal{V}). \tag{1}$$

VCMR can be decomposed into two subtasks: first *retrieve* the untrimmed videos v^* that contain the target moment m^* related to the given query, then *localize* target moment m^* from the retrieved video v^*. Following existing practice [11], [12], [13], [14], we apply an off-the-shelf search engine to retrieve the top-k videos with the highest similarity to the query from the video corpus. Then, we work on locating the best matching m^* from these candidate videos. The prediction of the target moment depends on the probabilities of start and end positions:

$$P(m|v^*, q) = P(t^s|v^*, q) \cdot P(t^e|v^*, q). \tag{2}$$

A. Feature Preparation

Given a query, we employ pre-trained RoBERTa [40] and standard Transformer [41], [42] to extract token features for each word in the query, *i.e.*, $\mathbf{Q} \in \mathbb{R}^{L\times d}$, where L is the length of the query. For each video, we use pre-trained SlowFast [43] and ResNet152 [44] to extract the frame features and max-pooling the frame features in a short clip (1.5 seconds duration), and the subtitle features are taken by max-pooling the token features extracted by RoBERTa in a clip. The two modalities of video features are then refined as $\mathbf{F} \in \mathbb{R}^{N\times d}$ and $\mathbf{S} \in \mathbb{R}^{N\times d}$ by Multi-Modal Transformer [45], where N is the number of clips in the video.

B. Query-Specific Context Learning

Each moment in the video is described by both visual frames and textual subtitles. As depicted in Fig. 3, we design a query-specific context learning module to capture modality-complementary and semantic-consistent multimodal contexts. Considering that each modality in the video exhibits distinct query relevance [14], such as action descriptions align more with visual elements, while character conversations are more related to textual information, we decompose the original query $\mathbf{Q} = \{\mathbf{q}_i\}_{i=1}^{L}$ into two modality-specific subqueries, $\mathbf{q}^s$ and $\mathbf{q}^f$, for subtitles and frames, respectively. Technically, we adopt additive attention to yield the subqueries:

$$\mathbf{w}^m = \text{Softmax}(\mathbf{W}^m\mathbf{Q}), \tag{3}$$

$$\mathbf{q}^m = \sum_i w_i^m \mathbf{q}_i \in \mathbb{R}^d, \tag{4}$$

where $m \in \{f, s\}$ indicates different modalities, and $\mathbf{W}^m$ is a learnable matrix.

With the modularized subqueries $\mathbf{q}^s$ and $\mathbf{q}^f$, we can capture relevant contexts from untrimmed videos and subtitles, respectively. Intuitively, a straightforward approach is to utilize cross-attentive mechanism to perform cross-modal interaction:

$$\mathbf{F}' = \text{CrossAtt}^f(\mathbf{F}, \mathbf{q}^f) \in \mathbb{R}^{N\times d}, \tag{5}$$

$$\mathbf{S}' = \text{CrossAtt}^s(\mathbf{S}, \mathbf{q}^s) \in \mathbb{R}^{N\times d}, \tag{6}$$

where CrossAtt$(\cdot)$ summarizes the cross-modal attention operation in [14] and [15]. Formally, given matrix $\mathbf{A}$ and vector $\mathbf{b}$, $\text{CrossAtt}(\mathbf{A}, \mathbf{b}) = \text{norm}(\mathbf{WA} \cdot \mathbf{b}) \odot \mathbf{WA}$, where $\mathbf{W}$ is a fully-connected layer, norm$(\cdot)$ represents L2-normalization, and $\odot$ denotes element-wise multiplication. Next, the visual and textual contexts can be combined as multimodal contexts via a linear layer. To capture the fine-grained contextual dependencies between video and query, we further apply bi-directional attention [46] on $\mathbf{F}'$, $\mathbf{S}'$ and $\mathbf{Q}$. It gives the query-specific multimodal context $\mathbf{C}$:

$$\mathbf{C} = \text{BiDAtt}\big(\text{Linear}([\mathbf{F}'; \mathbf{S}']), \mathbf{Q}\big) \in \mathbb{R}^{N\times d} \tag{7}$$

where [;] denotes row-wise concatenation. BiDAtt$(\cdot)$ contains query-to-video and video-to-query attentions as [46]. We omit the linear and transformer encoder layers for brevity.

Semantic Contrastive Loss: As mentioned before, one of the keys of VCMR is to obtain effective multimodal contexts for moment localization. A key criterion to achieve this goal is that the contexts derived from frames and subtitles are semantically consistent, *i.e.*, both are closely related to the query. Inspired by this, we introduce a semantic contrastive

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loss $\mathcal{L}^{sc}$ to penalize the contexts that include query-irrelevant semantics, thereby mitigating the risk of attention drift.

Specifically, we take the given query q as an anchor and construct a triple of $\{q, v^+, v^-\}$, where the positive video v^+ is the ground-truth video paired with the query. The negative video v^- is randomly sampled from the negative video pool of the query, constructed using the query-video similarity scores from the first-stage retrieval engine, following existing practices [11], [18].

For each video, we first calculate the unimodal contexts from frames and subtitles using Eq. (5), resulting in two context triples $\{\mathbf{Q}, \mathbf{F}'^+, \mathbf{F}'^-\}$ and $\{\mathbf{Q}, \mathbf{S}'^+, \mathbf{S}'^-\}$. Then, we calculate the sentence-clip semantic relevance between positive/negative video samples and query anchors. For instance, the relevance between the query $\bar{\mathbf{q}} \in \mathbb{R}^d$ and j-th clip $\mathbf{f}_j'^+ \in \mathbb{R}^d$ in the context $\mathbf{F}'^+$ is given as:

$$r_j^{f+} = \mathrm{Cos}(\bar{\mathbf{q}}, \mathbf{f}_j'^+), \tag{8}$$

where $\bar{\mathbf{q}}$ is the average pooling of the word sequence $\mathbf{Q} = \{\mathbf{q}_i\}_{i=1}^L$, computed as $\bar{\mathbf{q}} = \frac{1}{L}\sum_i \mathbf{q}_i$. $\mathrm{Cos}(\cdot)$ denotes cosine similarity. In a similar way, we can obtain the relevance scores r_j^{f-} for $(\mathbf{Q}, \mathbf{F}'^-)$, r_j^{s+} for $(\mathbf{Q}, \mathbf{S}'^+)$, and r_j^{s-} for $(\mathbf{Q}, \mathbf{S}'^-)$.

These relevance scores are normalized given both positive and negative relevance scores within a specific modality, *i.e.*, (r_j^{f+}, r_j^{f-}) or (r_j^{s+}, r_j^{s-}). We thus obtain $p^f = \sum_{j=1}^N (r_j^{f+})$ and $p^s = \sum_{j=1}^N (r_j^{s+})$ that represents the probability that the visual context $\mathbf{F}'^+$ and textual context $\mathbf{S}'^+$ not contain any negative semantics, respectively. The p^f and p^s increase with an increase in the positive relevance scores relative to the negative relevance scores. The semantic contrastive loss is formulated as:

$$\mathcal{L}^{sc} = -\sum_{v,q\in\mathcal{D}} (\log(p^f) + \log(p^s)), \tag{9}$$

where $\mathcal{D}$ is the training dataset. The objective $\mathcal{L}^{sc}$ encourages high relevance scores in positive instances, which means the visual $\mathbf{F}'^+$ and textual $\mathbf{S}'^+$ contexts consistently are related to the query $\mathbf{Q}$.

C. Progressive Moment Localization

With the query-specific multimodal context $\mathbf{C}$, the remaining research problem is to predict accurate start/end boundaries of the target moment (*i.e.*, two video frames) from a large number of candidate videos. To address this challenge, we propose a progressive moment localization method as shown in Fig. 3. At first, we introduce two region labels, namely foreground and background. We label the frames within the target moment and adjacent L frames as the foreground, while the remaining frames as background. Our motivation lies in the observation that real-world video moments often have fuzzy boundaries where adjacent frames still hold important contexts. Including these frames may help the model handle boundary uncertainties and improve localization accuracy. Accordingly, a region predictor is used to predict the probability $\mathbf{P}^{reg}$ of a clip being foreground/background:

$$\mathbf{P}^{reg} = \mathrm{Softmax}\big(\mathcal{R}^{cls}(\mathbf{C})\big) \in \mathbb{R}^{N\times 2}, \tag{10}$$

where $\mathcal{R}^{cls}$ is the region predictor that comprises two 1D-convolution layers with ReLU activation [47]. Note that during the forward propagation, we apply softmax; whereas in backpropagation, we employ gumbel-softmax [48] to ensure the smooth flow of gradients.

Then, we introduce orthogonal region embeddings $\mathbf{E}^{reg} \in R^{2\times d}$ to represent the region labels, which are randomly initialized according to the orthogonality criterion and tuned during end-to-end optimization. After region label predictions, we can infuse label information into multimodal contexts $\mathbf{C}$ and obtain the label-attended multimodal contexts $\mathbf{C}'$ as:

$$\mathbf{C}' = \mathbf{P}^{reg}\mathbf{E}^{reg} + \mathbf{C}. \tag{11}$$

Superimposed with the region embeddings, $\mathbf{C}'$ contains rich priors about the foreground and background frames, which provide useful guidance for the model to accurately detect the moment boundaries in subsequent localization tasks.

Following [12], we respectively feed $\mathbf{C}'$ into a bidirectional GRU (BiGRU) layer [35] and a 1D convolution (Conv1D) layer [5] to generate the start probability $\mathbf{P}^{start}$ and end probability $\mathbf{P}^{end} \in \mathbb{R}^N$ over all frames. The BiGRU is to capture temporal dependencies within $\mathbf{C}'$, enabling comprehensive learning from past and future contexts. The Conv1D layer then maps contexts to output probabilities. Formally,

$$\mathbf{P}^{start} = \mathrm{Softmax}(\mathrm{Conv1D}(\mathrm{BiGRU}(\mathbf{C}'))), \tag{12}$$

$$\mathbf{P}^{end} = \mathrm{Softmax}(\mathrm{Conv1D}(\mathrm{BiGRU}(\mathbf{C}'))). \tag{13}$$

Region-aware Loss: In our progressive localization approach, the region predictions $\mathbf{P}^{reg}$ and region embeddings $\mathbf{E}^{reg}$ are critical components that provide essential foreground and background cues for the localizer. To further improve localization accuracy, we introduce a region-aware loss $\mathcal{L}^{ra}$. On one hand, we employ a cross-entropy loss to supervise the region predictor, thereby enhancing the model's ability to discern video regions. On the other hand, we enforce orthogonality in the learnable region embeddings to prevent positional information confusion between the foreground and background regions. The $\mathcal{L}^{ra}$ is formulated as:

$$\mathcal{L}^{ra} = \mathrm{CE}(\mathbf{P}^{reg}, \mathbf{Y}^{reg}) + ||\mathbf{E}^{reg\top}\mathbf{E}^{reg} \odot (\mathbf{1} - \mathbf{I})||^2, \tag{14}$$

where $\mathrm{CE}(\cdot)$ represents the cross-entropy loss function and $\mathbf{Y}^{reg}$ denotes the ground truth region labels. $\mathbf{1}$ is the matrix with all elements being 1 and $\mathbf{I}$ is the identity matrix. The second term in Eq. (14) is the orthogonal regularization [49], which ensures $\mathbf{E}^{reg}$ to keep the orthogonality.

D. Learning

In addition to $\mathcal{L}^{sc}$ and $\mathcal{L}^{ra}$, our model is also constrained with the basic moment localization loss $\mathcal{L}^{ml}$ for VCMR. $\mathcal{L}^{ml}$ is based on cross-entropy loss:

$$\mathcal{L}^{ml} = \mathrm{CE}(\mathbf{P}^{start}, \mathbf{Y}^{start}) + \mathrm{CE}(\mathbf{P}^{end}, \mathbf{Y}^{end}), \tag{15}$$

where $\mathbf{Y}^{start}$ and $\mathbf{Y}^{end}$ are one-hot labels for the start and end timestamps of the ground truth moment, respectively.

Total Loss: Our final model is trained with all three aforementioned objectives:

$$\mathcal{L} = \lambda_{ml}\mathcal{L}^{ml} + \lambda_{sc}\mathcal{L}^{sc} + \lambda_{ra}\mathcal{L}^{ra}, \tag{16}$$

where λ_* are hyperparameters to balance the three losses.

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IV. Experiment

A. Experimental Setup

1) Datasets: We evaluate our method on two datasets: TVR [10] and DiDeMo [50]. The **TVR** dataset is a multimodal retrieval dataset based on 6 long-running TV shows. It includes around 21.8K videos with timestamped subtitles and 109K queries. The average video duration is about 76.2 seconds and the average moment length is about 9.1 seconds. The query contains about 13.4 words on average and each video has 5 queries. Following [11], we split 17,435/2,179/1,089 videos for train/validation/test, respectively. The **DiDeMo** dataset [50] is an open-world dataset built on YFCC100M [51]. It contains over 40K pairs of video moments and corresponding descriptions. The average video duration is around 54 seconds and the average moment duration is 6.5 seconds. Each video has 4 queries and the query contains about 8 tokens. We adopt the standard split [11] of: 8,395 videos for training, 1,065 videos for validation, and 1,004 videos for testing.

2) Metrics: Following previous studies [11], [12], [13], [19], we adopt "$R@k, IoU = m$" as the evaluation metric. This metric shows the percentage of testing queries having at least one correct retrieval (with IoU score larger than m) in the top-k retrieved moments.

3) Implementation Details: We use RoBERTa [40] to extract the text features of queries and subtitles. The feature dimension is 768-D. Visual features are extracted by SlowFast [43] and ResNet [44]. The dimension of the combined SlowFast+ResNet features is 4,352-D. For the Transformer layers used in QCLPL, we set the hidden size to 384 and the intermediate size to 1,536. The number of the attention head is 4. The hyper-parameters in Eq. (16) are set to $\lambda_{ml} = 0.01$, $\lambda_{sc} = 0.015$ and $\lambda_{ra} = 0.01$. Following [11], we set the number of clips for each moment candidate between 1∼24 and 3∼7 for TVR and DiDeMo datasets, respectively. Similar to [11], [12], and [13], we use HERO [37] as the video retriever. During training, we employ the video retriever to sample 10 and 2 negative videos as hard negatives for Shared-Norm [11] to train the moment localizer for TVR and DiDeMo, respectively. We use the AdamX optimizer with a learning rate $1e$-4. The batch size is set to 32 by default for all datasets. We use the early stop schedule that the model will stop when the sum of R@1, IoU = {0.5, 0.7} exceeds 5 epochs without promotion.

B. Comparison With Existing VCMR Methods

Tables I and II present the comparisons with the existing VCMR methods on the TVR and DiDeMo datasets, respectively. Our QCLPL outperforms all the competitor models with clear margins on both datasets. There are two observations in Table I: (1) QCLPL shows the best localization performance on all metrics, especially on R@1. For example, compared with the latest method DMFAT [13], we achieve a relative improvement of 37.8% in R@1, IoU = 0.5 and 34.8% in R@1, IoU = 0.7. Previous methods often ignore the localization complexity posed by the large and hard retrieval space, and directly apply a 1D convolutional layer to predict the moment boundary. By introducing the progressive localization strategy, the performance is greatly improved. (2) The multimodal fusion in existing methods (*e.g.*, CKCN) using scalar weights involves redundant and query-independent contexts. In contrast, our method constructs query-specific multimodal context and shows superior advantages, improving the R@100, IoU = 0.7 from 39.87 reported by CKCN to 40.97.

TABLE I
Results of VCMR on the TVR Validation Set

Method	Venue	R@k, IoU = 0.5			R@k, IoU = 0.7		
		R@1	R@10	R@100	R@1	R@10	R@100
XML [10]	ECCV'20	-	-	-	2.62	9.05	22.47
FLAT [38]	ArXiv'20	8.45	21.14	30.75	4.61	11.29	16.24
HAMMER [38]	ArXiv'20	9.19	21.28	31.25	5.13	11.38	16.71
HERO [37]	EMNLP'20	-	-	-	5.13	16.26	24.55
ReLoCLNet [19]	SIGIR'21	8.03	21.37	44.10	4.15	14.06	32.42
CONQUER [11]	ACM MM'21	13.68	33.75	46.23	7.76	22.49	35.17
MPGN [52]	EMNLP'22	-	-	-	6.49	19.12	38.33
SAN [53]	ICASSP'22	-	-	-	7.03	20.24	38.97
SQuiDnet [18]	ECCV'22	-	-	-	8.52	-	-
CTDL [39]	ICASSP'23	-	-	-	8.03	-	-
CKCN [12]	TMM'23	13.69	30.73	47.94	7.92	22.00	39.87
DMFAT [13]	TCVST'23	13.75	34.10	47.97	7.99	23.81	36.86
QCLPL	**Ours**	**18.95**	**38.07**	**50.34**	**10.77**	**29.21**	**40.97**

TABLE II
Results of VCMR on the DiDeMo Testing SET

Method	Venue	R@k, IoU = 0.5			R@k, IoU = 0.7		
		R@1	R@5	R@10	R@1	R@5	R@10
XML [10]	ECCV'20	2.26	-	10.42	1.59	-	6.77
HERO [37]	EMNLP'20	3.37	8.79	13.26	2.76	7.73	11.78
ReLoCLNet [19]	SIGIR'21	2.28	-	-	1.71	-	-
CONQUER [11]	ACM MM'21	3.31	9.27	13.99	2.79	8.04	11.9
SAN [53]	ICASSP'22	-	-	-	2.76	-	13.56
SQuiDnet [18]	ECCV'22	-	-	-	2.62	-	10.28
CTDL [39]	ICASSP'23	-	-	-	3.21	-	12.73
CKCN [12]	TMM'23	3.22	9.01	13.10	2.69	7.62	11.08
T-CKCN [12]	TMM'23	4.06	10.14	14.18	3.44	8.97	12.60
DMFAT [13]	TCVST'23	3.44	9.79	14.01	2.89	8.71	12.41
PREM [14]	ICMR'24	3.51	-	-	2.98	-	-
QCLPL	**Ours**	**4.42**	**11.73**	**16.00**	**3.62**	**9.89**	**13.64**

The results in Table II consistently demonstrate the superiority of our method on DiDeMo. Moreover, we find that the performance of all models on the DiDeMo dataset is worse than that on the TVR dataset, which is mainly due to the insufficient labeled data in the DiDeMo dataset [12].

To this end, T-CKCN [12] proposed to transfer the knowledge learned on TVR to the training process on DiDeMo. Without extra knowledge, our method still outperforms T-CKCN by mining semantically complementary multimodal contexts from subtitles and frame streams.

C. Evaluation on SVMR

To fully evaluate our proposed QCLPL, we conduct experiments on the SVMR subtask. The results are reported in Table III. SVMR task aims to localize the target moments

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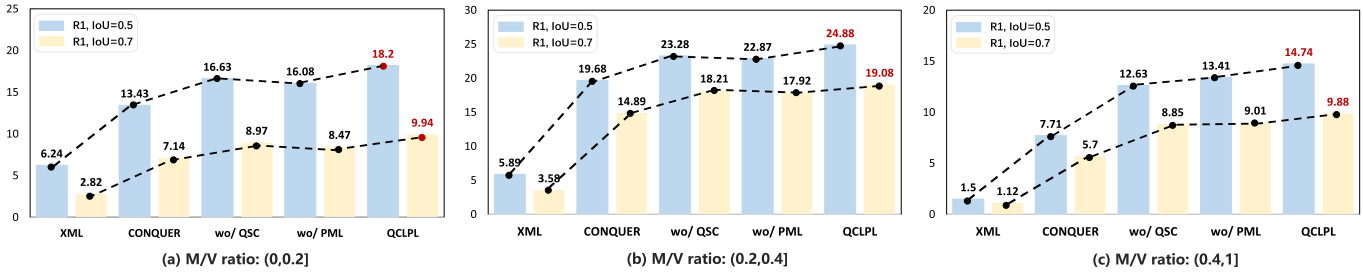

Fig. 4. M/V performance comparison on TVR. Queries are grouped according to their moment-to-video ratios $r(M/V)$, *i.e.*, the proportion of the query-related moment in the entire video. The smaller $r(M/V)$ indicates more challenging queries.

TABLE III
RESULTS OF SVMR SUBTASK ON TVR AND DIDEMO DATASETS. $*$ INDICATES OUR REPRODUCTION BY OFFICIAL CODE

Dataset	Method	Venue	R@1 , IoU=μ	
			μ=0.5	μ=0.7
TVR	XML [10]	ECCV'20	30.75	13.89
	HERO [37]	EMNLP'20	-	15.30
	TAN* [54]	AAAI'20	15.07	7.49
	Moment-DETR* [55]	NeurIPS'21	37.34	18.95
	ReLoCLNet [19]	SIGIR'21	31.88	15.04
	CONQUER [11]	ACM MM'21	42.57	22.84
	SQuiDnet [18]	ECCV'22	41.31	24.74
	CKCN [12]	TMM'23	43.38	23.18
	DMFAT [13]	TCVST'23	43.51	23.26
	PREM [14]	ICMR'24	46.29	24.83
	QCLPL	**Ours**	**47.30**	**25.68**
DiDeMo	HERO [37]	EMNLP'20	39.20	30.19
	ReLoCLNet [19]	SIGIR'21	34.81	26.71
	CONQUER [11]	ACM MM'21	38.17	29.90
	CKCN [12]	TMM'23	36.54	28.99
	PREM [14]	ICMR'24	40.79	33.77
	QCLPL	**Ours**	**40.83**	**34.38**

in a given video. As reported, our QCLPL outperforms comparison methods on both datasets, including two advanced approaches specifically designed for the SVMR task, TAN [54] and MomentDETR [55]. This is attributed to the proposed progressive moment localization(PML) method, which allows the model to focus on searching for target moments within the foreground while mitigating the distraction caused by query-independent background.

D. Ablation Studies

In this section, we present ablation experiment results to verify the effectiveness of the important components and design choices of our proposed model. The results are presented in Tables IV∼VI and Figs. 4∼5.

1) Main Components: From Table IV (#1 and #3), we can first confirm that progressive moment localization is important for reducing background distractions in the localizer for both SVMR and VCMR tasks. On the TVR dataset, introducing PML brings an 8.84% and 8.88% relative improvement at R@1, IoU = 0.5 under the VCMR and SVMR settings, respectively. Additionally, we conduct experiment (#2) by replacing our PML with 1D convolutional layers as [10], [11], [12], [13], and [14]. For the VCMR task on TVR, the R@1,IoU = 0.5 metric drops from 18.95 (experiment #7) to 16.98, clearly showing the crucial role of PML in addressing the localization challenges in VCMR. Comparing experiments #3 and #4, the learned query-specific contexts **C** can clearly improve the moment retrieval performances, where the R@1, IoU = 0.5 on VCMR increases from 17.12 to 17.62, which is further enhanced to 18.35 by applying modal-specific subqueries in experiment #6. This proves that subqueries really contribute to capturing useful information for multi-modal contexts. Additionally, semantic contrastive loss $\mathcal{L}^{sc}$ and region-aware loss $\mathcal{L}^{ra}$ are important to ensure the effectiveness of QSC and PML modules. Thus as shown by comparing experiments #5, #6, and #7, with gradually adding $\mathcal{L}^{sc}$ and $\mathcal{L}^{ra}$, QCLPL achieves the best results. These ablations demonstrate the effectiveness of our designed components in improving moment retrieval performance.

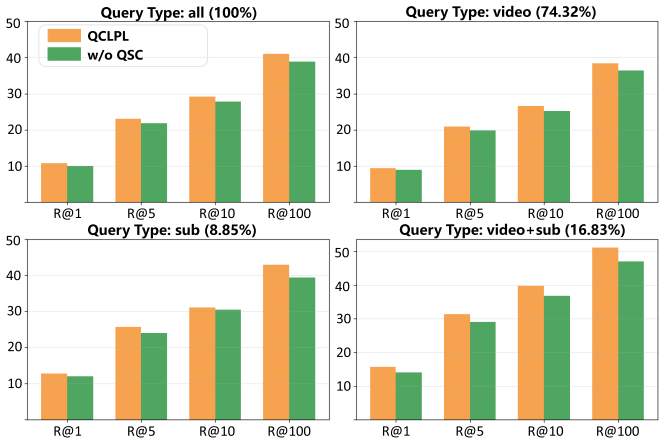

Fig. 5. Performance on different types of queries on TVR. Each query is labeled based on its relevance to either the video, the subtitle, or both.

2) Robustness to Different Query Types: On TVR dataset, queries are categorized into three types indicating that are more relevant to the video, subtitles, or both [10]. To validate the generalizability of the QSC module to various query types, we conduct experiments on the three subsets as shown in Fig. 5. w/o QSC is implemented by experiment #3 in Table IV. As observed, QCLPL markedly outperforms the w/o QSC across all types of queries, especially for queries related to video+subtitle. The learned query-specific contexts **C** effectively integrates useful and complementary semantics in the subtitle and frame streams related to given queries.

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TABLE IV

ABLATION STUDIES ON THE TVR VALIDATION SET. ∗ REPRESENTS AN ALTERNATIVE IMPLEMENTATION OF QSC WHERE THE QUERY IS NOT DECOMPOSED INTO TWO MODALITY-SPECIFIC SUBQUERIES, AND THE ORIGINAL QUERY IS USED DIRECTLY. **QSC**: QUERY-SPECIFIC MULTIMODAL CONTEXT **C**. **PML**: PROGRESSIVE MOMENT LOCALIZATION. $\mathcal{L}^{ra}$: REGION-AWARE LOSS. $\mathcal{L}^{sc}$: SEMANTIC CONTRASTIVE LOSS

Exp.	QSC	PML	L^{ra}	L^{sc}	TVR (val)				DiDeMo (test)			
					VCMR		SVMR		VCMR		SVMR	
					IoU=0.5	IoU=0.7	IoU=0.5	IoU=0.7	IoU=0.5	IoU=0.7	IoU=0.5	IoU=0.7
1	✗	✗	✗	✗	15.73	8.47	42.02	22.40	3.11	2.46	32.7	26.8
2	✓	✗	✗	✓	16.98	9.36	45.11	23.68	3.36	2.97	37.87	30.74
3	✗	✓	✓	✗	17.12	9.89	45.75	24.39	3.29	2.84	36.06	30.01
4	✓*	✓	✓	✗	17.62	10.19	46.25	24.70	3.57	3.01	38.37	32.55
5	✓	✓	✗	✗	17.93	10.08	45.79	24.87	3.73	3.12	39.64	33.53
6	✓	✓	✓	✗	18.35	10.35	46.84	25.18	4.12	3.35	40.34	33.91
7	✓	✓	✓	✓	**18.95**	**10.77**	**47.3**	**25.68**	**4.42**	**3.62**	**40.83**	**34.38**

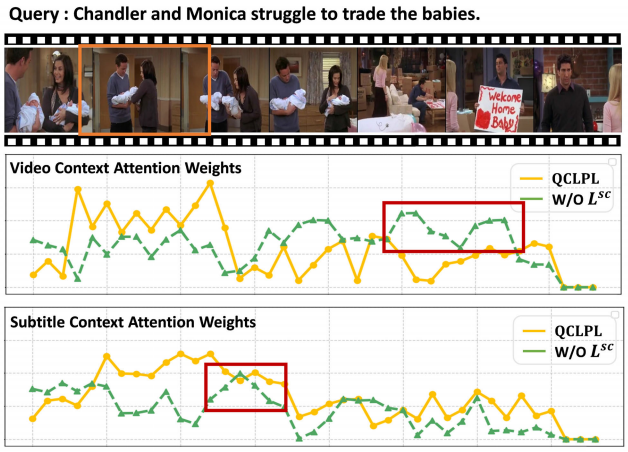

Fig. 6. Visualization of the cross-attention weights between the query and frames/subtitles over time, comparing the "w/o $\mathcal{L}^{sc}$" model and QCLPL. The red box indicates the attention drift in traditional temporal attention.

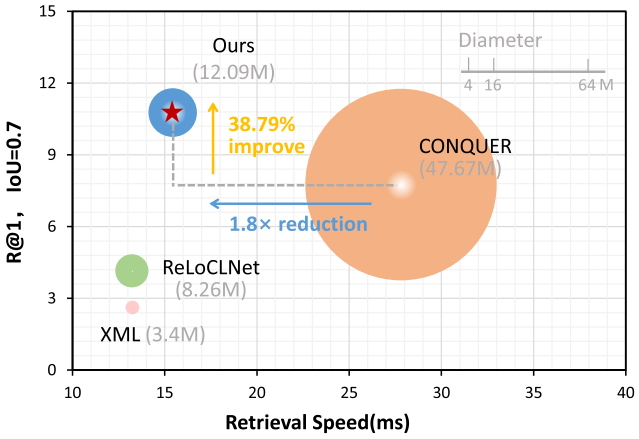

Fig. 7. The performance (*i.e.*, R@1,IoU=0.7), Speed(ms), and #parameters for various VCMR models on the TVR dataset. The center of the bubble indicates the value of R@1,IoU=0.7. The diameter of the bubble or star is proportional to the #parameters (M) while the horizontal axis indicates the speed (ms).

Moreover, we group the test queries based on their moment-to-video ratio $r(M/V)$, *i.e.*, the proportion of the query-related moments in the entire video. The smaller $r(M/V)$ indicates less relevant content while more irrelevant content in the target video with respect to the query, indicating that the corresponding query is more challenging. As shown in Fig. 4, we split TVR into three groups according to $r(M/V)$, where the moments are short ($r \in (0, 0.2]$), medium ($r \in (0.2, 0.4]$), and long ($r \in (0.4, 1.0]$). Our proposed model consistently performs the best, which again verifies its effectiveness. It is also worth noting that for short moments (Fig. 4 (a)), the PML module brings a greater performance improvement than the learned QSC, while for long moments (Fig. 4 (c)), the learned QSC contributes more. This is consistent with our solution. Shorter moments mean more redundant backgrounds, and PML can effectively reduce the interference of background for localization. Longer moments mean that the video contains rich context related to the target moment, and our query-specific context learning can help to mine multimodal context related to the query.

3) Issue of Attention Drift: Fig. 6 illustrates the cross-attention weights between the query and frames/subtitles over time, comparing the "w/o $\mathcal{L}^{sc}$" model and QCLPL. The "w/o $\mathcal{L}^{sc}$" model represents a variant where the query-specific context learning degenerates to traditional temporal attention. As shown in Fig. 6, the red box indicates the attention drift phenomenon in temporal attention, where irrelevant moments are incorrectly emphasized. In contrast, QCLPL effectively mitigates attention drift and focuses on query-relevant content. This demonstrates the importance of semantic contrastive learning in constructing effective multimodal contexts. This observation potentially explains why there is a noticeable performance drop in Table IV (#6 and #7).

In addition, we evaluate the similarity between queries and different contexts with and without $\mathcal{L}^{sc}$. We introduce a coarse-grained Embedding Matching-based score (EMScore) [56], which measures the cosine similarity between vision and text embeddings. For example, given the visual context $\mathbf{F}'$ and the query $\bar{\mathbf{q}}$, the metric $\text{EMScore}(\mathbf{F}', \bar{\mathbf{q}})_c$ is calculated as:

$$\text{EMScore}(\mathbf{F}', \bar{\mathbf{q}})_c = \frac{1}{N} \sum_{j=1}^{N} \mathbf{f}_j'^{\top} \bar{\mathbf{q}}, \tag{17}$$

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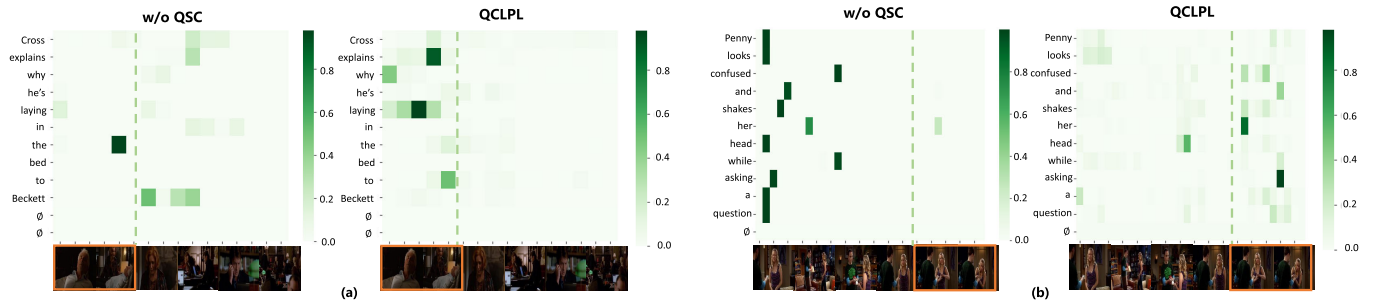

Fig. 8. Visualization of semantic similarity between the query and multimodal context, which are obtained by w/o QSC (*i.e.*, element-wise addition directly [11]) and our method, respectively.

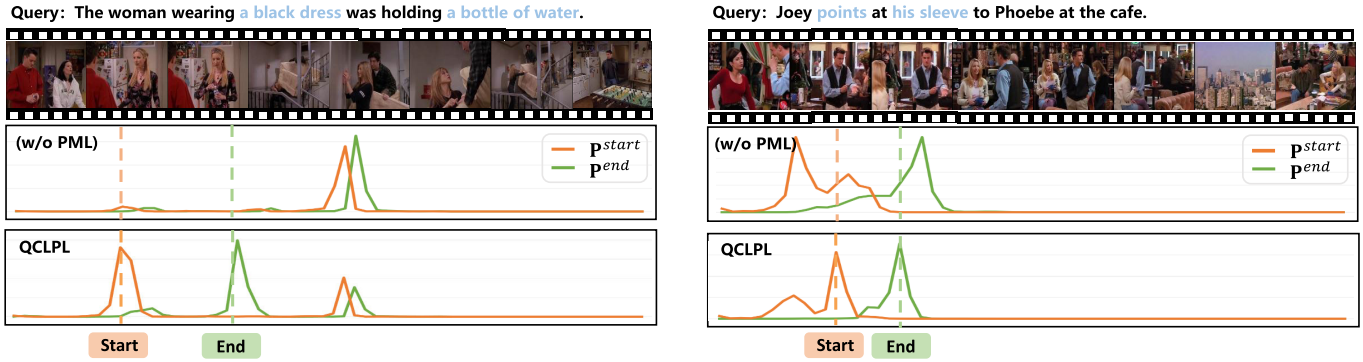

Fig. 9. Comparison of probabilities of start and end timestamps predicted by w/o PML and our QCLPL. The dashed lines indicate the ground-truth moment.

TABLE V
COMPARISON OF DIFFERENT METHODS BASED ON EMSCORE [56]

Method	EMScore$(\mathbf{F}', \bar{\mathbf{q}})_c$	EMScore$(\mathbf{S}', \bar{\mathbf{q}})_c$
w/o $\mathcal{L}^{sc}$	0.25	0.21
QCLPL	**0.32**	**0.35**

where all the embeddings are L2 normalized, the cosine similarity is reduced to the inner product. As shown in Table V, compared with the model "w/o $\mathcal{L}^{sc}$", introducing semantic contrastive loss significantly increases the similarity between the extracted context and the query.

4) Effect of Hyperparameters: The extension of foreground boundaries serves as a crucial setting in progressive moment localization. Generally, increasing the extension length L can introduce more contextual semantic information, yet it also leads to more background that interferes with the precise localization of target boundaries. To find a better trade-off, we investigate the model's performance when $L = \{0, 1, 2, 3, 4\}$. As shown in Table VI, as L increases, the performance first improves, then saturates and peaks with $L = 1$.

We also investigate the sensitivity of our model to the loss coefficients λ on the TVR dataset. Starting with a fixed $\lambda_{ml} = 0.01$, we vary λ_{sc} and λ_{ra} over the values $\{0.005, 0.01, 0.015, 0.03, 0.05\}$. As shown in Table. VII, our model maintains robust performance across a range of hyperparameter values, with the optimal trade-off achieved at $\lambda_{sc} = 0.015$ and $\lambda_{ra} = 0.01$ on TVR dataset. Each loss contributes to the retrieval performance, so keeping them within a similar order of magnitude ensures a balance between progressive localization and contextual learning.

TABLE VI
THE EFFECTS OF EXTENDING THE LENGTH OF FOREGROUND BOUNDARIES L. LARGER L HELPS TO INTRODUCE MORE CONTEXTUAL SEMANTIC INFORMATION, BUT IT ALSO LEADS TO MORE IRRELEVANT BACKGROUND. THE OPTIMAL VALUE OF L IS 1 FOR BOTH THE SVMR AND VCMR

Method	VCMR			SVMR	
	R@K, IoU=0.7			R@1, IoU=μ	
	R@1	R@10	R@100	μ=0.5	μ=0.7
L=0	10.38	28.85	40.23	46.83	25.17
L=1	**10.77**	**29.21**	**40.97**	**47.3**	**25.68**
L=2	10.46	29.08	40.31	46.92	25.34
L=3	10.15	28.58	39.27	46.44	24.96
L=4	10.07	28.25	39.46	46.26	24.78

E. Model Efficiency

In Fig. 7, we compare some open-source and competitive models mentioned above in terms of retrieval speeds and model parameters. For a fair comparison, we only monitor time consumption for the ranking procedure. Compared to Conquer [11], we have fewer parameters (12.09M vs. 47.67M), faster speed (1.8× faster), and better performance (38.8% improvement on R@1, IoU = 0.7). Our approach is independent of the reuse of Transformer for densely modeling video features and achieves better performance by enhancing multimodal context modeling and introducing region position cues. This clearly demonstrates the effectiveness and efficiency of our proposed QCLPL.

F. Qualitative Analysis

1) Similarity of Query and Context: As shown in Fig. 8, we provide two examples to illustrate the effectiveness of the

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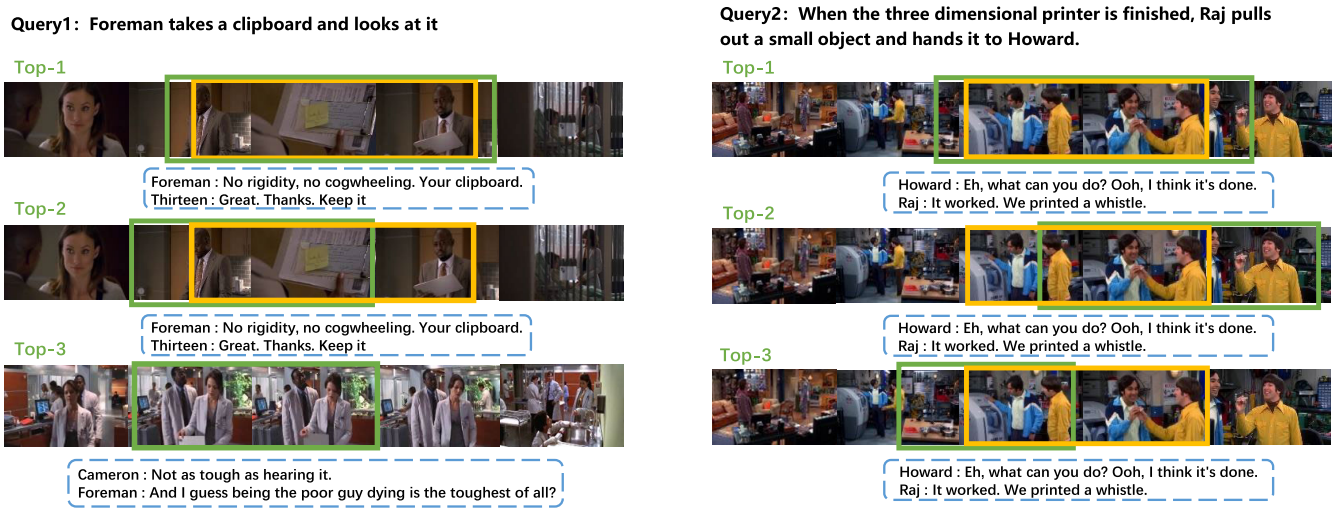

Fig. 10. VCMR results predicted by our method on TVR. We show top-3 retrieved moments for each query. Text inside dashed boxes is the subtitles associated with the predicted moments. Green and yellow boxes indicate the predictions and ground truth, respectively.

TABLE VII

EFFECTS OF THE HYPERPARAMETERS λ_{ml}, λ_{sc}, AND λ_{ra} ON TVR DATASET. λ_{ml} IS FIXED TO 0.01 FOR FAIR COMPARISONS. THE BEST PEAKS AT $\lambda_{sc} = 0.015$ AND $\lambda_{ra} = 0.01$

Item	Value	VCMR IoU=0.7				SVMR R@1	
		R@1	R@5	R@10	R@100	IoU=0.5	IoU=0.7
λ_{sc}	0.005	10.00	22.16	28.66	40.11	46.74	25.10
	0.01	10.35	22.55	28.76	40.54	47.01	25.36
	0.015	**10.77**	**23.07**	**29.21**	**40.97**	**47.3**	**25.68**
	0.03	10.38	22.79	28.93	40.63	47.10	25.41
	0.05	9.86	22.04	28.45	39.96	46.39	4.89
λ_{ra}	0.005	10.17	22.64	28.69	40.26	46.82	25.18
	0.01	**10.77**	**23.07**	**29.21**	**40.97**	**47.3**	**25.68**
	0.015	10.29	22.83	28.85	40.39	46.95	25.29
	0.03	9.78	22.39	28.27	39.87	46.43	24.93
	0.05	9.65	22.16	28.12	39.62	46.31	24.78

learned query-specific contexts (QSC), where the heatmap plots the cosine similarity between the query and multi-modal context integrated by frames and subtitles. It can be seen that the w/o QSC model fails to extract query-related multimodal information and inevitably exhibits wrong attention to video frames in background regions. However, our method successfully captures the relevant moments and shows deep understanding of the query. For example, the key semantics are captured such as "*laying in the bed*" in Fig. 8 (a) and "*shakes her head*" in Fig. 8 (b).

2) Prediction of Moment Boundaries: In Fig. 9, we further investigate how our proposed PML module guides the model to search the boundaries of target moment. As shown in Fig. 9 (left), there is a stark prediction deviation in w/o PML, while our QCLPL disregards irrelevant distractions and obtains more accurate localization results. In Fig. 9 (right), although the predicted boundary of w/o PML contains the target moment, it incorrectly introduces the adjacent irrelevant regions. By introducing progressive moment localization, our model effectively enhances the video features in the foreground region while mitigating the influence of background features unrelated to the query, thereby improving localization results.

3) VCMR Results: Fig. 10 shows our VCMR results on the TVR validation set for two given queries. As shown, by comprehensively understanding both the visual frames and textual subtitles, and employing progressive localization on untrimmed videos, we successfully retrieve highly relevant video moments from the video corpus, and rank those most relevant to the ground-truth moment as the top-1 candidate. Notably, in Query 1, the top-3 video does not contain the ground-truth moment, our model still retrieves it and finds a relevant moment describing "*takes a clipboard*". This demonstrates the exceptional ability to understand and discriminate fine-grained semantics in video content.

V. CONCLUSION

This paper proposes a novel VCMR method with Query-specific Context Learning and Progressive Localization (QCLPL), aiming to address existing challenges of modal heterogeneity and large search space. Specifically, we design a query-specific context learning module to mine modality-complementary and semantically consistent multimodal contexts from frames and subtitles. Furthermore, we introduce a semantic contrastive loss $\mathcal{L}^{sc}$ to penalize contexts that incorporate irrelevant semantics. To tackle the moment localization in a large search space, we propose a progressive localization method that facilitates boundary detection with the foreground and background region priors. We introduce a region-aware loss $\mathcal{L}^{ra}$, which operates in conjunction with a moment localization loss $\mathcal{L}^{ml}$ and $\mathcal{L}^{sc}$ for end-to-end optimization. Extensive experiments on two large datasets, TVR and DiDeMo, demonstrate the effectiveness and superiority of our proposed QCLPL.

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REFERENCES

[1] P. Song, D. Guo, X. Yang, S. Tang, and M. Wang, "Emotional video captioning with vision-based emotion interpretation network," *IEEE Trans. Image Process.*, vol. 33, pp. 1122–1135, 2024.

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[2] P. Song, D. Guo, X. Yang, S. Tang, E. Yang, and M. Wang, "Emotion-prior awareness network for emotional video captioning," in *Proc. 31st ACM Int. Conf. Multimedia*, Oct. 2023, pp. 589–600.

[3] P. Song, D. Guo, J. Cheng, and M. Wang, "Contextual attention network for emotional video captioning," *IEEE Trans. Multimedia*, vol. 25, pp. 1858–1867, 2023.

[4] P. Song et al., "Efficiently gluing pre-trained language and vision models for image captioning," *ACM Trans. Intell. Syst. Technol.*, vol. 15, no. 6, pp. 1–16, Dec. 2024.

[5] X. Yang, T. Chang, T. Zhang, S. Wang, R. Hong, and M. Wang, "Learning hierarchical visual transformation for domain generalizable visual matching and recognition," *Int. J. Comput. Vis.*, vol. 132, no. 11, pp. 4823–4849, Nov. 2024.

[6] X. Yang, J. Zeng, D. Guo, S. Wang, J. Dong, and M. Wang, "Robust video question answering via contrastive cross-modality representation learning," *Sci. China Inf. Sci.*, vol. 67, no. 10, pp. 1–16, Oct. 2024.

[7] W. Moon, S. Hyun, S. Park, D. Park, and J.-P. Heo, "Query–dependent video representation for moment retrieval and highlight detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 23023–23033.

[8] Y. Liu, S. Li, Y. Wu, C. W. Chen, Y. Shan, and X. Qie, "UMT: Unified multi-modal transformers for joint video moment retrieval and highlight detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 3042–3051.

[9] V. Escorcia, M. Soldan, J. Sivic, B. Ghanem, and B. Russell, "Finding moments in video collections using natural language," 2019, *arXiv:1907.12763*.

[10] J. Lei, L. Yu, T. L. Berg, and M. Bansal, "TVR: A large-scale dataset for video-subtitle moment retrieval," in *Proc. 16th Eur. Conf. Comput. Vis.*, 2020, pp. 447–463.

[11] Z. Hou, C.-W. Ngo, and W. K. Chan, "CONQUER: Contextual query-aware ranking for video corpus moment retrieval," in *Proc. 29th ACM Int. Conf. Multimedia*, Oct. 2021, pp. 3900–3908.

[12] T. Chen, W. Wang, Z. Jiang, R. Li, and B. Wang, "Cross-modality knowledge calibration network for video corpus moment retrieval," *IEEE Trans. Multimedia*, vol. 26, pp. 3799–3813, 2024.

[13] X. Zhang, P. Zhao, J. Ji, X. Lu, and Y. Yin, "Video corpus moment retrieval via deformable multigranularity feature fusion and adversarial training," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 8, pp. 6686–6698, Aug. 2024.

[14] D. Hou, L. Pang, H. Shen, and X. Cheng, "Improving video corpus moment retrieval with partial relevance enhancement," in *Proc. Int. Conf. Multimedia Retr.*, May 2024, pp. 394–403.

[15] D. Hou, L. Pang, Y. Lan, H. Shen, and X. Cheng, "Multi-video moment ranking with multimodal clue," 2023, *arXiv:2301.13606*.

[16] Y. Xiao et al., "Bridging the gap: A unified video comprehension framework for moment retrieval and highlight detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 18709–18719.

[17] X. Yang, J. Dong, Y. Cao, X. Wang, M. Wang, and T.-S. Chua, "Tree-augmented cross-modal encoding for complex-query video retrieval," in *Proc. 43rd Int. ACM SIGIR Conf. Res. Develop. Inf. Retr.*, Jul. 2020, pp. 1339–1348.

[18] S. Yoon et al., "Selective query-guided debiasing for video corpus moment retrieval," in *Proc. Eur. Conf. Comput. Vis.*, Oct. 2022, pp. 185–200.

[19] H. Zhang et al., "Video corpus moment retrieval with contrastive learning," in *Proc. 44th Int. ACM SIGIR Conf. Res. Develop. Inf. Retr.*, Jul. 2021, pp. 685–695.

[20] J. Gao, C. Sun, Z. Yang, and R. Nevatia, "TALL: Temporal activity localization via language query," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2017, pp. 5267–5275.

[21] X. Sun, J. Gao, Y. Zhu, X. Wang, and X. Zhou, "Video moment retrieval via comprehensive relation-aware network," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 9, pp. 5281–5295, Sep. 2023.

[22] Z. Qi, Y. Yuan, X. Ruan, S. Wang, W. Zhang, and Q. Huang, "Collaborative debias strategy for temporal sentence grounding in video," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 11, pp. 10972–10986, Nov. 2024.

[23] D. Han, X. Cheng, N. Guo, X. Ye, B. Rainer, and P. Priller, "Momentum cross-modal contrastive learning for video moment retrieval," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 7, pp. 5977–5994, Jul. 2024.

[24] H. Zhang, A. Sun, W. Jing, L. Zhen, J. T. Zhou, and S. M. R. Goh, "Parallel attention network with sequence matching for video grounding," in *Proc. Findings Assoc. Comput. Linguistics (ACL-IJCNLP)*, 2021, pp. 776–790.

[25] X. Yang, S. Wang, J. Dong, J. Dong, M. Wang, and T.-S. Chua, "Video moment retrieval with cross-modal neural architecture search," *IEEE Trans. Image Process.*, vol. 31, pp. 1204–1216, 2022.

[26] X. Yang, F. Feng, W. Ji, M. Wang, and T.-S. Chua, "Deconfounded video moment retrieval with causal intervention," in *Proc. 44th Int. ACM SIGIR Conf. Res. Develop. Inf. Retr.*, Jul. 2021, pp. 1–10.

[27] X. Sun, H. Wang, and B. He, "MABAN: Multi-agent boundary-aware network for natural language moment retrieval," *IEEE Trans. Image Process.*, vol. 30, pp. 5589–5599, 2021.

[28] W. Wang, Y. Huang, and L. Wang, "Language-driven temporal activity localization: A semantic matching reinforcement learning model," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 334–343.

[29] Y. Wang, W. Zhou, and H. Li, "Fine-grained semantic alignment network for weakly supervised temporal language grounding," in *Proc. Findings Assoc. Comput. Linguistics, EMNLP*, 2021, pp. 89–99.

[30] S. Kim, J. Cho, J. Yu, Y. Yoo, and J. Y. Choi, "Gaussian mixture proposals with pull-push learning scheme to capture diverse events for weakly supervised temporal video grounding," in *Proc. AAAI Conf. Artif. Intell.*, Mar. 2024, pp. 2795–2803.

[31] M. Zheng, Y. Huang, Q. Chen, Y. Peng, and Y. Liu, "Weakly supervised temporal sentence grounding with Gaussian-based contrastive proposal learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 15555–15564.

[32] M. Zheng, Y. Huang, Q. Chen, and Y. Liu, "Weakly supervised video moment localization with contrastive negative sample mining," in *Proc. AAAI Conf. Artif. Intell.*, 2022, pp. 3517–3525.

[33] Z. Wang, J. Chen, and Y.-G. Jiang, "Visual co-occurrence alignment learning for weakly-supervised video moment retrieval," in *Proc. 29th ACM Int. Conf. Multimedia*, Oct. 2021, pp. 1459–1468.

[34] H. Li et al., "D3G: Exploring Gaussian prior for temporal sentence grounding with glance annotation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 13734–13746.

[35] H. Zhang, A. Sun, W. Jing, and J. T. Zhou, "Temporal sentence grounding in videos: A survey and future directions," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 8, pp. 10443–10465, Aug. 2023.

[36] Y. Zhao, Z. Gao, C. Ma, W. Guan, R. Wang, and S. Chen, "Fine-grained modality relation-aware network for video moment retrieval," *IEEE Trans. Circuits Syst. Video Technol.*, early access, Nov. 18, 2024, doi: 10.1109/TCSVT.2024.3494744.

[37] L. Li et al., "HERO: Hierarchical encoder for video+ language omni-representation pre-training," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP) Workshops*, 2020, pp. 2046–2065.

[38] B. Zhang et al., "A hierarchical multi-modal encoder for moment localization in video corpus," 2020, *arXiv:2011.09046*.

[39] S. Yoon et al., "Counterfactual two-stage debiasing for video corpus moment retrieval," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Jun. 2023, pp. 1–5.

[40] Y. Liu et al., "RoBERTa: A robustly optimized BERT pretraining approach," 2019, *arXiv:1907.11692*.

[41] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Jun. 2017, pp. 5998–6008.

[42] S. Li, X. Xu, X. Jiang, F. Shen, X. Liu, and H. T. Shen, "Multi-grained attention network with mutual exclusion for composed query-based image retrieval," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 4, pp. 2959–2972, Apr. 2024.

[43] C. Feichtenhofer, H. Fan, J. Malik, and K. He, "SlowFast networks for video recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2019, pp. 6202–6211.

[44] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[45] V. Gabeur, C. Sun, K. Alahari, and C. Schmid, "Multi-modal transformer for video retrieval," in *Proc. 16th Eur. Conf. Comput. Vis.*, Glasgow, U.K., 2020, pp. 214–229.

[46] M. Seo, A. Kembhavi, A. Farhadi, and H. Hajishirzi, "Bidirectional attention flow for machine comprehension," 2016, *arXiv:1611.01603*.

[47] X. Glorot, A. Bordes, and Y. Bengio, "Deep sparse rectifier neural networks," in *Proc. 14th Int. Conf. Artif. Intell. Statist.*, 2011, pp. 315–323.

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[48] E. Jang, S. Gu, and B. Poole, "Categorical reparametrization with Gumble–Softmax," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Apr. 2017, pp. 1–12.
[49] A. Brock, J. Donahue, and K. Simonyan, "Large scale GAN training for high fidelity natural image synthesis," in *Proc. Int. Conf. Learn. Represent.*, Jan. 2018, pp. 1–15.
[50] L. A. Hendricks, O. Wang, E. Shechtman, J. Sivic, T. Darrell, and B. Russell, "Localizing moments in video with natural language," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 5803–5812.
[51] B. Thomee et al., "YFCC100M: The new data in multimedia research," *Commun. ACM*, vol. 59, no. 2, pp. 64–73, Jan. 2016.
[52] M. Jung, S. Choi, J. Kim, J.-H. Kim, and B.-T. Zhang, "Modal-specific pseudo query generation for video corpus moment retrieval," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2022, pp. 7769–7781.
[53] D. Kim, S. Yoon, J. W. Hong, and C. D. Yoo, "Semantic association network for video corpus moment retrieval," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2022, pp. 1720–1724.
[54] S. Zhang, H. Peng, J. Fu, and J. Luo, "Learning 2D temporal adjacent networks for moment localization with natural language," in *Proc. AAAI Conf. Artif. Intell.*, 2020, pp. 12870–12877.
[55] J. Lei, T. L. Berg, and M. Bansal, "Detecting moments and highlights in videos via natural language queries," in *Proc. Adv. Neural Inf. Process. Syst.*, Dec. 2021, pp. 11846–11858.
[56] Y. Shi et al., "EMScore: Evaluating video captioning via coarse-grained and fine-grained embedding matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 17929–17938.

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